Let SELFREJECT = { <M> : M rejects itself }.

Theorem: SELFREJECT is undecidable.

Proof: Suppose not. Let SR be a decider for SELFREJECT.

ACCEPT(SR) = SELFREJECT

DIVERGE(SR) = $\emptyset$

SR accepts <M> iff M rejects <M>

SR accepts <SR> iff SR rejects <SR>

SR accepts <X> iff SR rejects <X>

p iff ¬p contradiction

but this is a contradiction!

So SR cannot exist. So SELFREJECT is undecidable. □

diagonalization